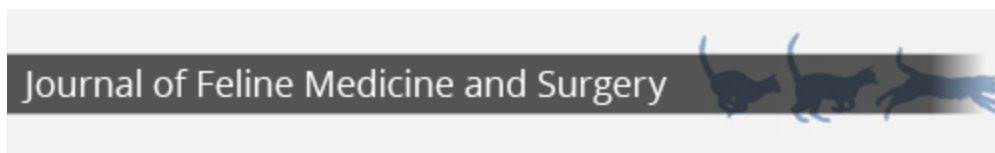


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Renoprotective effects of docosahexaenoic acid in cats with early chronic kidney disease due to polycystic kidney disease: a pilot study

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Abstract

Objectives

Lipids containing n-3 fatty acids have been reported to have protective effects on renal function, with docosahexaenoic acid (DHA) expected to be particularly effective. However, no reports have demonstrated the renoprotective effects of DHA-enriched lipids in cats with chronic kidney disease (CKD). Therefore, the aim of this pilot study was to examine the renoprotective effects of DHA-enriched fish oil in cats.

Methods

Five healthy cats and five cats with early non-azotaemic CKD due to autosomal dominant polycystic kidney disease (PKD) were orally administered DHA-enriched fish oil in liquid form (250 or 500 mg/kg body weight [BW] and 250 mg/kg BW of DHA, respectively) for 28 days.

Inappropriately dilute urine and markedly increased urinary *N*-acetyl-D-glucosamine (NAG) index were detected in cats with PKD before DHA-enriched fish oil administration. Changes in the fatty acid composition ratio in the blood of all 10 cats were assessed after orally administering 250 mg/kg of DHA.

Results

Post-administration, no adverse clinical effects were observed, and blood and urine tests were within the reference intervals in healthy cats. Cats with PKD showed significantly decreased serum symmetric dimethylarginine (SDMA), urine protein:creatinine ratio (UPC) and urinary NAG index at post-administration. Furthermore, oral administration of DHA-enriched fish oils significantly decreased the blood concentration ratio of arachidonic acid (AA) in cats with PKD post-administration. Furthermore, the concentration ratio of DHA in the blood significantly increased in both healthy cats and cats with PKD, and the DHA:AA ratio also increased.

Conclusions and relevance

Oral administration of DHA-enriched fish oils for 28 days significantly decreased blood AA levels and significantly increased DHA concentration and DHA:AA ratios in cats with PKD, and improved the SDMA, UPC and urinary NAG index, suggesting its potential for renoprotective effects in cats with early non-azotaemic CKD due to PKD.

Keywords: Potential renoprotective effects, docosahexaenoic acid, polycystic kidney disease, DHA, early non-azotaemic CKD

Introduction

Studies on the physiological functions of docosahexaenoic acid (DHA) and eicosapentaenoic acid (EPA) – the n-3 fatty acids – have been ongoing since 1970. Various useful bioactivities of DHA/EPA, including neutral lipid reduction and cognitive function improvement, and antithrombotic and anti-inflammatory effects, have been reported.¹⁻⁴ DHA is a very important body component found in many organs such as the brain, eye retina, heart, placenta and testes.⁵ As DHA and EPA are not easily biosynthesised in the body, they are considered ‘essential fatty acids’ that must be obtained through the diet. The daily requirement for polyunsaturated fatty acids (DHA and EPA combined) in cats is 60 mg/kg during the growth period and 1.6 mg/kg during the maintenance period.^{6,7}

One bioactivity of n-3 fatty acids is protecting renal function in humans and small animals.⁸⁻¹¹ The effects of feeding three different lipid sources (fish oil [high in n-3 fatty acids], beef tallow and safflower oil [high in n-6 fatty acids]) were observed in 15/16 nephrectomised dogs for 20 months.¹⁰ The results showed that the glomerular filtration rate of the fish oil-fed group was 1.5- and 2.2-times higher than that of the beef fat- and safflower oil-fed group, respectively, indicating that n-3 fatty acid-enriched lipids are expected to have a renoprotective effect. In recent years, technology that focuses on extracting high concentrations of DHA and EPA, respectively, has been established.¹²⁻¹⁴ As fish oil enriched with EPA or DHA has become available, differences in the effects of DHA and EPA on renal function have been reported. For example, a study that assessed the effects of highly purified DHA and EPA or EPA-only administration on rat renal function in an animal model of metabolic syndrome showed that the use of combination therapy with DHA and EPA improved or prevented renal failure.¹⁵ Halade et al¹⁶ reported that, in a rat model of lupus nephritis caused by systemic lupus erythematosus, rats treated with DHA-fortified oil (EPA 5%/DHA 60%), but not EPA-fortified oil (EPA 55%/DHA 5%), had a significantly prolonged lifespan (DHA-fortified group: 848 days; EPA-fortified group: 658 days), significantly lower serum dsDNA antibody levels and renal IgG deposition, and lower urinary protein levels. Furthermore, our previous study reported that the treatment of salt-loaded, stroke-prone, spontaneously hypertensive rats (SHR-SP) with DHA-enriched fish oil (70%) suppressed stroke onset and blood pressure elevation, protected renal glomeruli and renal blood vessels and reduced proteinuria.¹⁷ These findings suggest that DHA, but not EPA, may be the most potent n-3 fatty acid, with nephroprotective effects in various animal species.

Although products fortified with 'n-3 fatty acids' have already been marketed in feline renal disease treatment diets, no reports have verified the effects of the oral administration of lipids enriched with DHA in cats with chronic kidney disease (CKD). There has been one report on the effect of EPA on renal function in patients with human autosomal dominant polycystic kidney disease (PKD),¹⁸ but no reports of DHA-enriched fish oil administered to cats with PKD caused by *PKD1* mutation. Therefore, this study aimed to confirm the safety of fish oil enriched with DHA in healthy cats (study A) and to verify the protective effects of DHA on renal function in cats with early CKD due to PKD (study B).

Materials and methods

Animals

Study A

Five healthy mixed-breed laboratory cats (aged 4–10 years; three castrated males and two spayed females) were used in this study to evaluate the safety of DHA-enriched fish oil for cats ([Table 1](#); first and second tests). All cats were negative for *PKD1* mutation.

Table 1.

Population characteristics and renal function parameters of cats enrolled in the study before the administration of docosahexaenoic acid

	RI	First test Healthy (n = 5)	Second test Healthy (n = 5)	PKD (n = 5)
Age (years)	–	7 (6–9)	7 (6–9)	6 (2–6)
Body weight (kg)	–	4.0 (3.1–4.5)	4.0 (3.0–4.2)	2.5 (1.9–4.2)
BCS (/5)	–	3 (3–4)	3 (3–4)	3 (2–3)
Serum creatinine (mg/dl)	0.80–1.50	1.11 (0.80–1.24)	1.48 (1.21–1.64)	0.91 (0.87–1.49)
BUN (mg/dl)	11–30	28.5 (25.7–28.6)	25.1 (23.3–26.8)	17.6 (17.5–21.2)
SDMA (mg/dl)	0–14	NA	NA	12.0 (11.0–12.0)
USG	1.035–1.050	1.041 (1.038–1.047)	1.038 (1.030–1.039)	1.017 (1.011–1.027)
UPC	<0.4	0.1 (0.09–0.19)	0.15 (0.10–0.15)	0.11 (0.10–0.19)

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Data are presented as median (interquartile range)
RI = reference interval; PKD = polycystic kidney disease; BCS = body condition score; BUN = blood urea nitrogen; SDMA = serum symmetric dimethylarginine; NA = not applicable; USG = urine specific gravity; UPC = urine protein:creatinine ratio

Five laboratory cats with stage 1 CKD (aged 2–9 years; three intact females and one spayed female [cats 1–4]) or stage 2 (aged 6 years; one spayed female [cat 5]) based on International Renal Interest Society (IRIS) staging were used to examine the protective effects of DHA-enriched fish oil on renal function ([Table 1](#)). Three intact females were Exotic Shorthairs and two spayed females were Himalayans; two of the three Exotic Shorthairs were very small and weighed 1.8–1.9 kg (body condition score 3/5). All cats were positive for *PKD1* mutation; ultrasound showed multiple cysts in both kidneys. These cats had blood tests, urinalysis and ultrasound examinations performed every 3 months for 1 year prior to the study. No significant clinical signs were observed in cats with PKD, excluding polyuria.

Cats in studies A and B were maintained at 24°C under a 12 h/12 h light/dark photoperiod (lights on at 7.00 am). All cats were individually housed in cages. The diet for cats was the same for 6 months prior to the start of the study and for its duration. Healthy cats were fed a commercial dry food for maintenance, and cats with PKD were fed a renal diet twice daily. The amount of food given to each cat was adequate for its daily nutritional needs. All procedures followed local and national animal ethics guidelines, and were approved by the animal research committee of Iwate University (approval numbers A201905 and A201906).

Oral administration of DHA-enriched fish oils

Fish oils enriched with DHA in liquid form (DHA-RS; Maruha Nichiro) were orally administered. The composition ratios of major unsaturated fatty acids were 7% n-6 and 78% n-3 fatty acids. Of these, 70% were DHA and 4% were EPA. The weight composition of DHA was 58% w/w. To prevent lipid oxidation, DHA-enriched fish oils were stored at –20°C before use.

Study A

DHA-enriched fish oils were orally administered at a dose of 250 mg/kg body weight (BW) as DHA q24h for 28 days before the evening feeding. The fish oil was mixed with a small amount of liquid food and fed to the cats in a free-feeding manner. After ensuring that the cat licked up all the fish oil, the cat was given dry food. After a 2-week washout period, cats were orally administered 500 mg/kg of DHA q24h for 28 days. During the administration period, the cats were allowed to drink water freely and were given common food in the morning and evening.

Study B

DHA-enriched fish oils were orally administered with DHA at a dosage of 250 mg/kg BW q24h for 28 days before the evening feeding in the same manner as in study A. During the administration period, animals were allowed to drink water freely and given a renal diet (feline renal support dry food; Royal Canin) in the morning and evening.

Sample collection and analysis

General physical examination and confirmation of DHA-enriched fish oil intake were performed daily for the entire study period. Fasting time was the same for all cats and all time points. Fasting blood and urine samples were collected on days 0 (baseline) and 28 of all studies. Blood samples were collected into lithium heparinised tubes, tubes with sodium citrate added and K₂-EDTA tubes. A complete blood count was performed with EDTA anticoagulated whole blood using a haematology analyser (pocH-100i; Sysmex). Cats were screened with a serum biochemical panel (glucose, blood urea nitrogen, creatinine, total protein, albumin, total cholesterol [TC], triglyceride [TG], total bilirubin, alanine aminotransferase [ALT], aspartate aminotransferase [AST], alkaline phosphatase, gamma-glutamyltransferase, calcium, inorganic phosphorus [iP], magnesium, sodium, potassium and chloride) using a biochemical autoanalyser (Accute TBA-40FR; Toshiba); coagulation tests (prothrombin time [PT] and activated partial thromboplastin time [APTT]) using a blood coagulation analysis system (COAG2NV; A&T); and blood fatty acid composition (arachidonic acid [AA] and DHA) using gas chromatography with AOAC international method 996.06.¹⁹ Serum symmetric dimethylarginine (SDMA) was analysed using an external laboratory (IDEXX Laboratories; Japan) with a high-throughput immunoassay. Alpha-1-acidic glycoprotein (α 1-AG) was measured only in cats with PKD and analysed by latex agglutination method using an external laboratory (FUJIFILM VET Systems; Japan).

Urine was collected via ultrasound-guided cystocentesis into uncoated tubes, processed within 30 mins of collection and analysed with dipstick testing and urine sediment testing. Furthermore, the following urinary biomarkers were determined: urine specific gravity (USG), urine protein:creatinine ratio (UPC), fractional excretion of electrolyte (FE Na, FE K, FE Cl) and *N*-acetyl-D-glucosamine (NAG) for cats with PKD. The method of Sato et al²⁰ was used to measure urinary NAG. Briefly, urinary NAG activity was measured using 5 mM *p*-nitrophenyl *N*-acetyl-beta-D-glucosaminide as a substrate and a spectrometer at 405 nm. Urinary NAG activity was expressed as U/l, and the NAG index was calculated using the following equation: NAG index (U/g) = urinary NAG activity (U/l)/urinary creatinine concentration (g/l). The normal feline reference interval (RI) of urinary NAG index is 2.2 ± 3.2 U/g.

Statistical analysis

For all data, the median (interquartile range [IQR]) was calculated. Data between baseline and DHA administration were compared using Wilcoxon's signed-rank test for all data. A *P* value <0.05 was considered to be statistically significant. Bell curve for Microsoft Excel version 4.02 was used for the statistical analyses (Social Survey Research Information).

Results

Study A: evaluation of the safety of DHA-enriched fish oil in healthy cats

All cats ingested all DHA-enriched fish oil in study A. The mean BWs of healthy cats receiving 250 mg of oral DHA remained unchanged at 4.02 kg (IQR 3.08–4.48) and 3.86 kg (IQR 3.1–4.32) at days 0 and 28 of treatment, respectively. After DHA administration, body weight increased by 0.12 kg in one cat and decreased by 0.12 kg to 0.16 kg in four cats. Other general physical examination findings were unchanged and no clinical problems were revealed. When 500 mg/kg DHA was orally administered, BW remained unchanged at 3.98 kg (IQR 2.98–4.16) and 3.86 kg (IQR 3.10–4.32) at days 0 and 28, respectively. After DHA administration, body weight increased by 0.02 kg to 0.04 kg in the two cats, and decreased by 0.06 kg to 0.22 kg in the three. However, 4/5 cats exhibited soft stools for a short period. Throughout the entire period, the median period of observed soft stools at 250 mg/kg of DHA and 500 mg/kg of DHA was 5 days (IQR 3–6) and 4 days (IQR 4–5), respectively. At 500 mg/kg of DHA, feeding preference was reduced in 4/5 cats due to the increased amount of fish oil. The cat with the highest amount of leftover dry food showed a decrease in food intake of 60% for 1 day, 30–46% for 3 days and 6–14% for 5 days. Biochemical blood tests showed significant differences in decreased glucose, TC, TGs and magnesium, and increased ALT, AST, iP, magnesium and potassium; however, these changes were within the normal RI ([Table 2](#)). PT and APTT, indicators of the blood coagulation system, did not change pre- or post-administration of DHA, regardless of the dose. No significant changes in USG, UPC or FE were also observed ([Table 2](#); FE data not shown).

Table 2.

Effects of docosahexaenoic acid (DHA)-enriched fish oil administration on the blood and urine of five healthy cats

	RI	DHA 250 mg/kg BW		DHA 500 mg/kg BW	
		0 days	28 days	0 days	28 days
RBCs ($\times 10^6/\mu\text{l}$)	5.0–10.0	8.8 (8.5–9.0)	8.1 (7.4–8.2)	9.2 (7.6–9.9)	8.3 (7.5–9.2)
Platelets ($\times 10^3/\mu\text{l}$)	20–80	21 (18–37)	26 (15–37)	15 (14–31)	19 (9–35)
PT (s)	9.3–11.3	10.9 (10.3–12.7)	11.0 (10.5–11.8)	10.9 (10.7–10.9)	10.7 (9.0–11.3)
APTT (s)	28–42	67.9 (61.3–81.8)	70.7 (66.8–72.0)	54.9 (46.6–69.4)	39.9 (31.0–43.5)
Glucose (mg/dl)	68–128	83 (83–84)	69 (65–79) [*] _–	84 (68–90)	77 (75–95)
TC (mg/dl)	63–181	96 (70–104)	105 (78–110)	97 (78–112)	109 (106–121) [*] _–
TG (mg/dl)	21–66	47 (29–48)	21 (20–28)	14 (11–26)	7 (7–9) [*] _–
ALT (U/l)	6–60	59 (59–64)	77 (65–81) [*] _–	72 (70–74)	56 (53–59)
AST (U/l)	10–50	19 (19–24)	30 (25–35) [*] _–	28 (25–35)	24 (21–33)
BUN (mg/dl)	11–30	29 (26–29)	25 (23–17)	25 (23–27)	27 (25–28)
Creatinine (mg/dl)	0.8–1.5	1.1 (0.8–1.2)	1.1 (1.0–1.2)	1.4 (1.2–1.6)	1.6 (1.2–1.7)
iP (mg/dl)	3.9–6.8	4.3 (3.9–4.5)	5.1 (4.7–5.2) [*] _–	4.6 (4.4–4.6)	5.1 (4.3–5.1)
Magnesium (mg/dl)	1.8–2.9	2.9 (2.9–3.1)	2.1 (2.1–2.19)	2.9 (2.8–3.0)	2.3 (2.2–2.3) [*] _–
Potassium (mmol/l)	3.4–4.6	4.2 (4.1–4.2)	4.5 (4.4–4.5) [*] _–	4.4 (4.3–4.7)	4.2 (4.0–4.2)
USG	1.035–1.050	1.041 (1.038–1.047)	1.031 (1.031–1.040)	1.038 (1.030–1.039)	1.034 (1.031–1.036)

	RI	DHA 250 mg/kg BW		DHA 500 mg/kg BW	
		0 days	28 days	0 days	28 days
UPC	<0.4	0.10 (0.09–0.19)	0.11 (0.08–0.15)	0.15 (0.10–0.15)	0.11 (0.10–0.20)

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Data are presented as median (interquartile range)

* $P < 0.05$

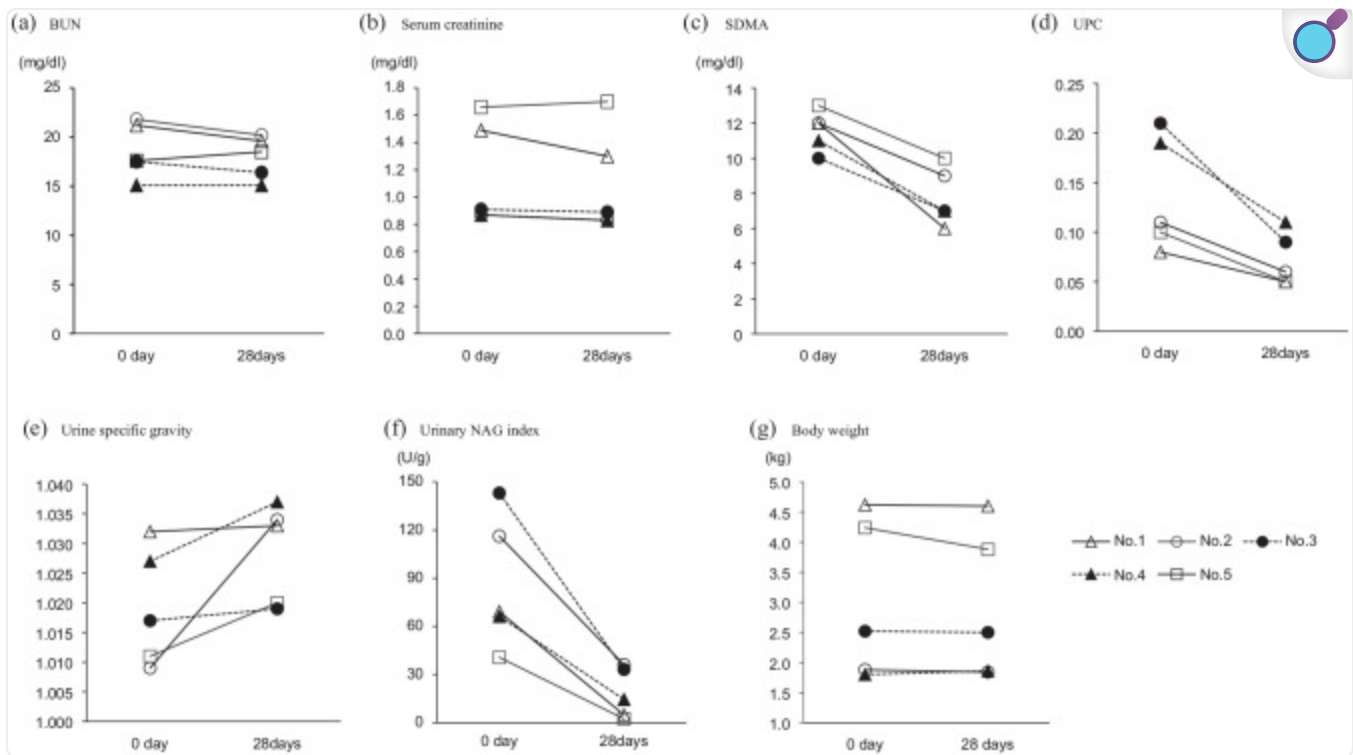
RI = reference interval; BW = body weight; RBCs = red blood cells; PT = prothrombin time; APTT = activated partial thromboplastin time; TC = total cholesterol; TG = triglycerides; ALT = alanine aminotransferase; AST = aspartate aminotransferase; BUN = blood urea nitrogen; iP = inorganic phosphorus; USG = urine specific gravity; UPC = urine protein:creatinine ratio

Study B: protective effects of the oral administration of DHA-enriched fish oil on renal function in cats with early CKD due to PKD

In cats with PKD, urinary concentration capacity decreased from the low specific urine gravity at baseline (1.017 [IQR 1.011–1.027]), and proximal tubular damage was observed from an elevated urinary NAG index (69.1 U/g [IQR 66.3–116.3]; RI 2.2 ± 3.2 U/g).

The median BWs of cats with PKD receiving 250 mg of DHA orally were 2.5 kg (IQR 1.9–4.2) and 2.7 kg (IQR 2.0–3.6) at days 0 and 28 of treatment, respectively ([Figure 1](#)). Other general physical examination findings were also unchanged, indicating no clinical problems. Blood tests showed significant variations in TC, AST, magnesium and sodium, which were all within their respective RIs ([Table 3](#)).

Figure 1.



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Effects on renal function and body weight of docosahexaenoic acid (DHA)-enriched fish oil administered orally to cats with polycystic kidney disease for 28 days. (a) Blood urea nitrogen; (b) serum creatinine; (c) symmetric dimethylarginine; (d) urine protein:creatinine ratio; (e) urine specific gravity; (f) urinary *N*-acetyl-D-glucosamine index; (g) body weight.

Line graphs represent individual data (Δ = cat 1; $\square$ = cat 2; $\bullet$ = cat 3; $\blacktriangle$ = cat 4; $\circ$ = cat 5)

Table 3.

Effects of docosahexaenoic acid (DHA)-enriched fish oil administration on the blood test results of five cats with polycystic kidney disease

	RI	DHA 250 mg/kg BW	
		0 days	28 days
WBCs (/μl)	5500–19,500	10,300 (3775–11,750)	11,300 (947–13,050)
Platelets (×10 ³ /μl)	200–800	252 (142–294)	238 (110–425)
α1-AG (μg/ml)	<736	277 (154–410)	249 (132–306)
TC (mg/dl)	63.0–181.0	99.4 (73.5–101.0)	125.7 (115.0–128.1)* _–
Triglycerides (mg/dl)	21.0–66.0	12.6 (11.1–16.5)	24.5 (24.2–32.3)
AST (U/l)	10.0–50.0	30.7 (22.7–120.8)	24.7 (21.7–30.2)* _–
Magnesium (mg/dl)	1.8–2.9	2.1 (2.0–2.2)	2.3 (2.2–2.3)* _–
Sodium (mmol/l)	147–156	152 (152–152)	154 (154–155)* _–

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Data are median (interquartile range)

**P* < 0.05

RI = reference interval; WBCs = white blood cells; α1-AG = alpha-1-acid glycoprotein; TC = total cholesterol; AST = aspartate aminotransferase

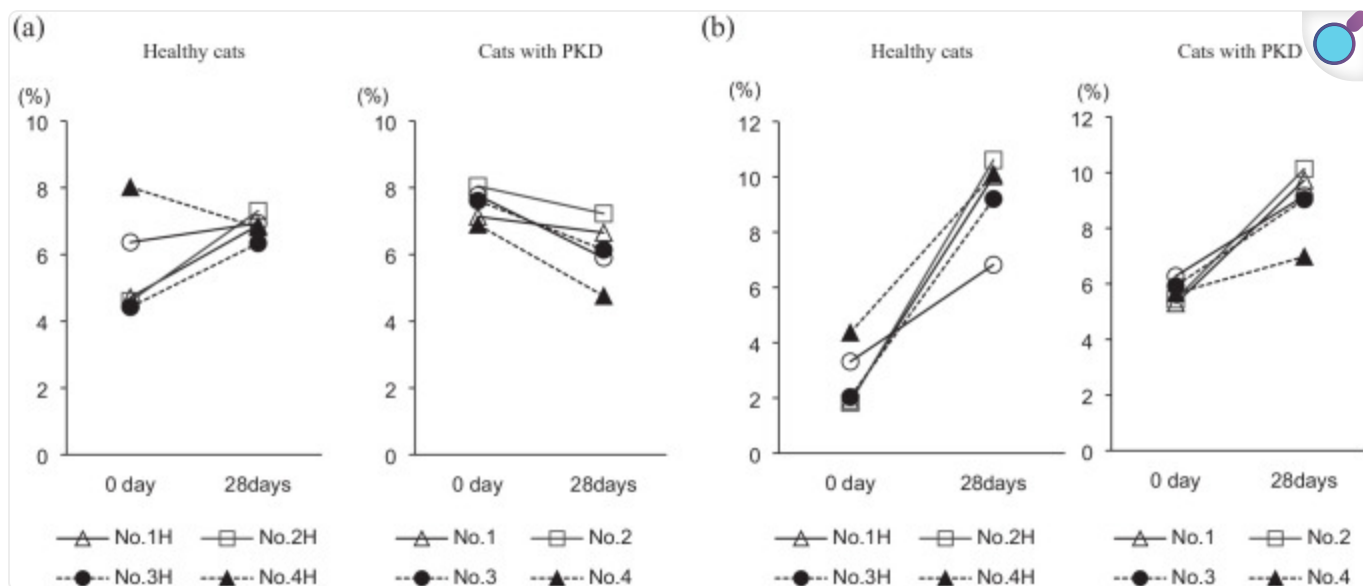
After 250 mg/kg DHA administration for 28 days, the renal function parameters of cats with PKD improved ([Figure 1](#)). A statistically significant decrease in SDMA, an index of glomerular filtration rate, from 12 μg/dl (IQR 11–12) to 7 μg/dl (IQR 7–9) at days 0 and 28 of treatment (*P* < 0.05), respectively, was observed in cats with PKD. Urinalysis showed a statistically significantly decreased UPC from 0.11 (IQR 0.10–0.19) to 0.06 (IQR 0.05–0.09; *P* < 0.05), and urinary NAG index from 69.1 U/g (IQR 66.3–116.3) to 14.5 U/g (IQR 4.8–33.2; *P* < 0.05) – a marker of acute proximal tubular injury – after the oral administration of 250 mg/kg DHA. USG increased from 1.017 (IQR 1.011–1.027) to 1.031 (IQR 1.022–1.034; *P* < 0.05). No urinary tract infections were seen in any cats

before or after DHA administration. Administration of 250 mg/kg DHA did not affect the FE of cats with PKD (data not shown).

Effects of oral administration of DHA-enriched fish oils on the fatty acid composition ratio in the blood of healthy cats and cats with PKD

Changes in the fatty acid composition ratio in the blood of healthy cats and cats with PKD post-oral DHA administration of 250 mg/kg for 28 days were investigated ([Figure 2](#)). The concentration ratio of AA in the blood was statistically significantly reduced by the administration of DHA in cats with PKD: 7.6% (IQR 7.0–7.9) and 6.1% (IQR 5.3–6.9) on days 0 and 28 of administration ($P < 0.05$), respectively. In both healthy and PKD-affected cats, the concentration ratio of DHA in the blood statistically significantly increased post-DHA administration ($P < 0.05$), indicating that the DHA concentration ratio in healthy cats was 2.0% (IQR 1.9–3.8) and 10% (IQR 8.0–10.4) on days 0 and 28 of administration, respectively, and that in PKD-affected cats it was 5.7% (IQR 5.4–6.1) and 9.2% (IQR 8.0–9.9) on days 0 and 28 of administration, respectively. Median DHA:AA ratios in healthy cats significantly increased from 0.46 (IQR 0.40–0.53) to 1.45 (IQR 1.22–1.47; $P < 0.05$). Median DHA:AA ratios for PKD cats also significantly increased from 0.78 (IQR 0.71–0.82) to 1.47 (IQR 1.42–1.51; $P < 0.05$). The percentages of AA and DHA in cats with PKD after the administration of 250 mg/kg DHA for 28 days showed little variability from day to day (data not shown).

Figure 2.



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Changes in the fatty acid composition ratio in the blood of healthy cats and cats with polycystic kidney disease (PKD) after the oral administration of 250 mg/kg docosahexaenoic acid (DHA) for 28 days. (a) Concentration ratio of arachidonic acid in the blood. (b) Concentration ratio of DHA in the blood.

Line graphs represent individual data (healthy cats: △ = cat 1H; □ = cat 2H; ● = cat 3H; ▲ = cat 4H; ○ = cat 5H; cats with PKD: △ = cat 1; □ = cat 2; ● = cat 3; ▲ = cat 4; ○ = cat 5)

Discussion

No clinical problems were observed in healthy cats administered 250 or 500 mg/kg DHA combined with food for 28 days, and blood and urine test data were within the RIs. The administration of n-3 fatty acids has been discussed with regard to platelet aggregation.^{21–23} In particular, EPA has an antithrombotic effect by antagonistically inhibiting the thromboxane A2 production, which has platelet-aggregating ability, and may contribute to bleeding as a side effect in rats and human.^{24,25} Saker et al²⁶ reported that the platelet aggregation of cats fed with foods containing different n-6 to n-3 fatty acid ratios was affected but not in a clinically significant manner. Bright et al²⁷ reported that a relatively high dose of n-3 fatty acids at approximately 1.8–

2.6 g daily containing EPA (1.1–1.7 g) and DHA (0.6–0.9 g) did not affect the indices involved in platelet aggregation and bleeding times in cats.²⁷ Sano et al²⁸ reported that PT and APTT in rats fed the fish oil diet were longer than for the rats that were fed the control diet. Conversely, eight clinical intervention studies in humans reported that long-chain polyunsaturated fatty acids of fish oil had no effect on blood coagulation parameters and bleeding manifestations.²⁹ In this study, DHA-enriched fish oil was administered, but no significant changes in platelet numbers, haemostasis time during blood collection, PT or APTT were observed in healthy cats. Based on the above results, DHA at a dose of ≤ 500 mg/kg is well tolerated in cats and can be administered for at least 1 month. However, some cats developed temporary soft stools; thus, DHA-enriched fish oil should be administered while monitoring the health status of cats.

Based on several other reports from humans, mice and rats,^{16,17,30} a minimum of approximately 375 mg/kg BW total fish oil can be administered to dogs and cats to lower proteinuria;^{31,32} however, this dose is extrapolated from human studies. Only one report actually verified the dosage of fish oils affecting the renal function of dogs and cats – the aforementioned model experiment in dogs¹⁰ – and a few reports have been published on cats. To the best of our knowledge, this study was the first to examine the oral administration of DHA-enriched fish oil in cats with IRIS stage 1 or 2 CKD on the background of *PKD1* mutation positivity. The results showed that DHA administration at 250 mg/kg for 28 days significantly decreased SDMA, UPC and urinary NAG, which are renal function parameters, compared with pre-administration. This suggests the potential protective effects of renal function at lower doses compared with doses that have been effective in dogs and cats.^{31,32}

Borderline proteinuria (UPC 0.2–0.4) has been reported as a risk factor for shortened survival in cats.³³ In this study, individuals with renal borderline proteinuria and those with a UPC close to 0.2 showed a decreased UPC after treatment with DHA-enriched fish oil, but, owing to the small sample size, we could not accurately assess cats with borderline proteinuria. In addition, as it has been reported that there is day-to-day variability in UPC in dogs,³⁴ the day-to-day variability in proteinuria or accuracy of the measurement of UPC could contribute to the changes in UPC when dealing with the low level of proteinuria. Further work is needed to evaluate this finding in cats.

DHA – an n-3 fatty acid – is converted into anti-inflammatory oxylipins (docosanoids) by oxidative enzymes (P450, lipoxygenase, cyclooxygenase, etc) in vivo. Conversely, AA – an n-6 fatty acid – is similarly converted by biological oxidases into oxylipins, some of which are hydroxylated fatty acids and prostaglandins, which increase inflammatory mediators such as transforming growth factor beta and tumour necrosis factor alpha. For example, 20-hydroxyeicosatetraenoic acid, produced by the action of P450 on AA, has been reported to elevate blood pressure and result in

inflammation, as well as increase growth factors for renal epithelial cells in PKD. ³⁵ In this study, DHA administration significantly increased blood DHA levels in PKD cats but significantly decreased the AA levels. It has been reported that increasing the DHA concentration in fish oil given to rats promoted tissue incorporation of DHA and EPA with a decrease in omega-6 AA, but whether an increase in blood DHA and a decrease in AA is necessary for renoprotective effects needs to be investigated in the future. ³⁶

From these findings, we speculated that docosanoids – the bioactive lipids converted from DHA by oxidative enzymes in vivo – may have antagonised inflammatory eicosanoids and suppressed inflammation, contributing to the renoprotective effect.

In our previous study, stroke-prone hypertensive rats generated by a high salt load of 8% demonstrated a significantly reduced albuminuria and increased glomerular filtration rate compared with the control group after the oral administration of ≥ 1.2 g/kg DHA for 1 month. ¹⁷ As a histopathological mechanism of renal function improvement, suppression of hyaline cast formation, glomerulosclerosis and renal artery damage in the kidneys of hypertensive rats by ingesting at least 1.2 g/kg DHA was found, proving the protective effects of DHA on glomeruli and renal arteries. Although histopathological studies were not performed in this study, the mechanism is thought to be the same as in rats, and further studies are needed to clarify the mechanism. Another mechanism is that DHA reduces oxidative stress in the kidney, which may alleviate tubular damage, leading to a significant decrease in urinary NAG and a trend toward an increase in USG, as shown in the present study. Furthermore, in this experiment, $\alpha 1$ -AG, an inflammation marker in cats, was not changed by the oral administration of DHA-enriched fish oils. This is consistent with the finding that DHA administration did not affect the acute inflammatory marker, C-reactive protein, in stroke-prone hypertensive rats. ¹⁷ As the pretreatment values of rats with CKD in this experiment were within the RIs, it was considered that DHA administration maintained the levels in this experiment due to the absence of systemic inflammation.

Our study had several limitations. First, the number of study subjects was low, and the cause of CKD was only PKD. Second, validation with long-term treatment studies (ie, >2 months) has not yet been conducted. Third, the study did not include a control group of cats with PKD that were given an alternative composition of DHA-free oil. Finally, the possibility that the changes identified in this study were caused by day-to-day variability in a small group of cats has not been ruled out. In the future, long-term effects, reproducibility and day-to-day variability in the administration of DHA in a large number of cats should be validated, and metabolite analysis should be performed to elucidate the nephroprotective mechanism of DHA.

Conclusions

We found that the oral administration of DHA-enriched fish oils significantly decreased blood AA levels but increased DHA concentration and DHA:AA ratios in cats with stage 1 or 2 CKD due to PKD. In this pilot study, the oral administration of DHA-enriched fish oils to cats with early CKD due to PKD improved SDMA, UPC and urinary NAG index, suggesting potential renoprotective effects of DHA.

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Footnotes

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References

1. Ikeda I, Wakamatsu K, Inayoshi A, et al. Alpha-linolenic, eicosapentaenoic and docosahexaenoic acids affect lipid metabolism differently in rats. *J Nutr* 1994; 124: 1898–1906. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
2. Nurk E, Drevon CA, Refsum H, et al. Cognitive performance among the elderly and dietary fish intake: the Hordaland Health Study. *Am J Clin Nutr* 2007; 86: 1470–1478. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
3. Heemskerk JWM, Vossen RCRM, van Dam-Mieras MCE. Polyunsaturated fatty acids and function of platelets and endothelial cells. *Curr Opin Lipidol* 1996; 7: 24–29. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
4. Calder PC. Dietary modification of inflammation with lipids. *Proc Nutr Soc* 2002; 61: 345–358. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
5. Hussein N, Fedorova I, Moriguchi T, et al. Artificial rearing of infant mice leads to n-3 fatty acid deficiency in cardiac, neural and peripheral tissues. *Lipids* 2009; 44: 685–702. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
6. National Research Council. Nutrient requirements and dietary nutrient concentrations. In: Kirk-Baer C, Schoen R. (eds). *Nutrient requirements of dogs and cats*. Washington, DC: The National Academies Press, 2006, pp 354–370. [[Google Scholar](#)]
7. Aii T. Fat and fatty acids necessary for dogs and cats [article in Japanese]. *Japanese Soc Pet Anim Nutr* 2006; 9: 13–23. [[Google Scholar](#)]
8. Syren ML, Turolo S, Marangoni F, et al. The polyunsaturated fatty acid balance in kidney health and disease: a review. *Clin Nutr* 2018; 37: 1829–1839. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
9. Hoogeveen EK, Geleijnse JM, Kromhout D, et al. Effect of omega-3 fatty acids on kidney function after myocardial infarction: the alpha omega trial. *Clin J Am Soc Nephrol* 2014; 9: 1676–1683. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
10. Brown SA, Brown CA, Crowell WA, et al. Beneficial effects of chronic administration of dietary omega-3 polyunsaturated fatty acids in dogs with renal insufficiency. *J Lab Clin Med* 1998; 131: 447–455. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

11. Brown SA, Brown CA, Crowell WA, et al. Effects of dietary polyunsaturated fatty acid supplementation in early renal insufficiency in dogs. J Lab Clin Med 2000; 135: 275–286. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)
12. Burja AM, Armenta RE, Radianingtyas H, et al. Evaluation of fatty acid extraction methods for *Thraustochytrium* sp. ONC-T18. J Agric Food Chem 2007; 55: 4795–4801. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)
13. Ozogul Y, Şimşek A, Balıkçı E, et al. The effects of extraction methods on the contents of fatty acids, especially EPA and DHA in marine lipids. Int J Food Sci Nutr 2012; 63: 326–331. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)
14. Señoráns M, Castejón N, Señoráns FJ. Advanced extraction of lipids with DHA from *Isochrysis galbana* with enzymatic pre-treatment combined with pressurized liquids and ultrasound assisted extractions. Molecules 2020; 25. DOI: 10.3390/molecules25143310. [\[DOI\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)
15. Katakura M, Hashimoto M, Inoue T, et al. Omega-3 fatty acids protect renal functions by increasing docosahexaenoic acid-derived metabolite levels in SHR.Cg-Leprcp/NDmcr rats, a metabolic syndrome model. Molecules 2014; 19. DOI: 10.3390/molecules19033247. [\[DOI\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)
16. Halade GV, Rahman MM, Bhattacharya A, et al. Docosahexaenoic acid-enriched fish oil attenuates kidney disease and prolongs median and maximal life span of autoimmune lupus-prone mice. J Immunol 2010; 184: 5280–5286. [\[DOI\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)
17. Kawarasaki M, Chiba Y, Kamata A, et al. Agents for the maintenance and protection of renal function, and methods for evaluating the effectiveness. Japanese Publication of Examined Patent Application or Granted Patent No. 7013416. 2022-01-21.
18. Higashihara E, Nutahara K, Horie S, et al. The effect of eicosapentaenoic acid on renal function and volume in patients with ADPKD. Nephrol Dial Transplant 2008; 23: 2847–2852. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)
19. AOAC International. AOAC fat (total, saturated and unsaturated) in foods, hydrolytic extraction gas chromatographic method. In: Horowitz W. (ed). AOAC official method 996.06. 18th ed. Arlington, VA: AOAC International, 2001. [\[Google Scholar\]](#)
20. Sato R, Soeta S, Syuto B, et al. Urinary excretion of *N*-acetyl- β -D-glucosaminidase and its isoenzymes in cats with urinary disease. J Vet Med Sci 2002; 64: 367–371. [\[DOI\]](#)

[\[PubMed\]](#) [\[Google Scholar\]](#)

21. Goodnight SH, Harris WS, Connor WE. The effects of dietary omega 3 fatty acids on platelet composition and function in man: a prospective, controlled study. *Blood* 1981; 58: 880–885. [\[PubMed\]](#) [\[Google Scholar\]](#)

22. Nordøy A, Davenas E, Ciavatti M, et al. Effect of dietary (n - 3) fatty acids on platelet function and lipid metabolism in rats. *Biochim Biophys Acta* 1985; 835: 491–500. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

23. Casali RE, Hale JA, LeNarz LR, et al. Improved graft patency associated with altered platelet function induced by marine fatty acids in dogs. *J Surg Res* 1986; 40: 6–12. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

24. Ding YA, Chang WK, Chen ML. Effects of dietary polyunsaturated fatty acid on platelet aggregation in rats. *Proc Natl Sci Counc Repub China B* 1989; 13: 102–108. [\[PubMed\]](#) [\[Google Scholar\]](#)

25. Dyerberg J, Bang HO, Stoffersen E, et al. Eicosapentaenoic acid and prevention of thrombosis and atherosclerosis? *Lancet* 1978; 312: 117–119. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

26. Saker KE, Eddy AL, Thatcher CD, et al. Manipulation of dietary (n-6) and (n-3) fatty acids alters platelet function in cats. *J Nutr* 1998; 128: 2645S–2647S. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

27. Bright JM, Sullivan PS, Melton SL, et al. The effects of n-3 fatty acid supplementation on bleeding time, plasma fatty acid composition, and in vitro platelet aggregation in cats. *J Vet Intern Med* 1994; 8: 247–252. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

28. Sano Y, Sato K, Uchida M, et al. Blood coagulation and fibrinolysis of rats fed fish oil: reduced coagulation factors especially involved in intrinsic pathway and increased activity of plasminogen activator inhibitor. *Biosci Biotechnol Biochem* 2003; 67: 2100–2105. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

29. Jeansen S, Witkamp RF, Garthoff JA, et al. Fish oil LC-PUFAs do not affect blood coagulation parameters and bleeding manifestations: analysis of 8 clinical studies with selected patient groups on omega-3-enriched medical nutrition. *Clin Nutr* 2018; 37: 948–957. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

30. De Caterina R, Caprioli R, Giannessi D, et al. n-3 fatty acids reduce proteinuria in patients with chronic glomerular disease. *Kidney Int* 1993; 44: 843–850. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
31. Harley L, Langston C. Proteinuria in dogs and cats. *Can Vet J* 2012; 53: 631–638. [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
32. Vaden SL, Elliott J. Management of proteinuria in dogs and cats with chronic kidney disease. *Vet Clin North Am Small Anim Pract* 2016; 46: 1115–1130. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
33. Syme HM, Markwell PJ, Pfeiffer D, et al. Survival of cats with naturally occurring chronic renal failure is related to severity of proteinuria. *J Vet Intern Med* 2006; 20: 528–535. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
34. LeVine DN, Zhang D, Harris T, et al. The use of pooled vs serial urine samples to measure urine protein:creatinine ratios. *Vet Clin Pathol* 2010; 39: 53–56. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
35. Park F, Sweeney WE, Jia G, et al. 20-HETE mediates proliferation of renal epithelial cells in polycystic kidney disease. *J Am Soc Nephrol* 2008; 19: 1929–1939. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
36. Miralles-Pérez B, Méndez L, Nogués MR, et al. Effects of a fish oil rich in docosahexaenoic acid on cardiometabolic risk factors and oxidative stress in healthy rats. *Mar Drugs* 2021; 19: 555. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]